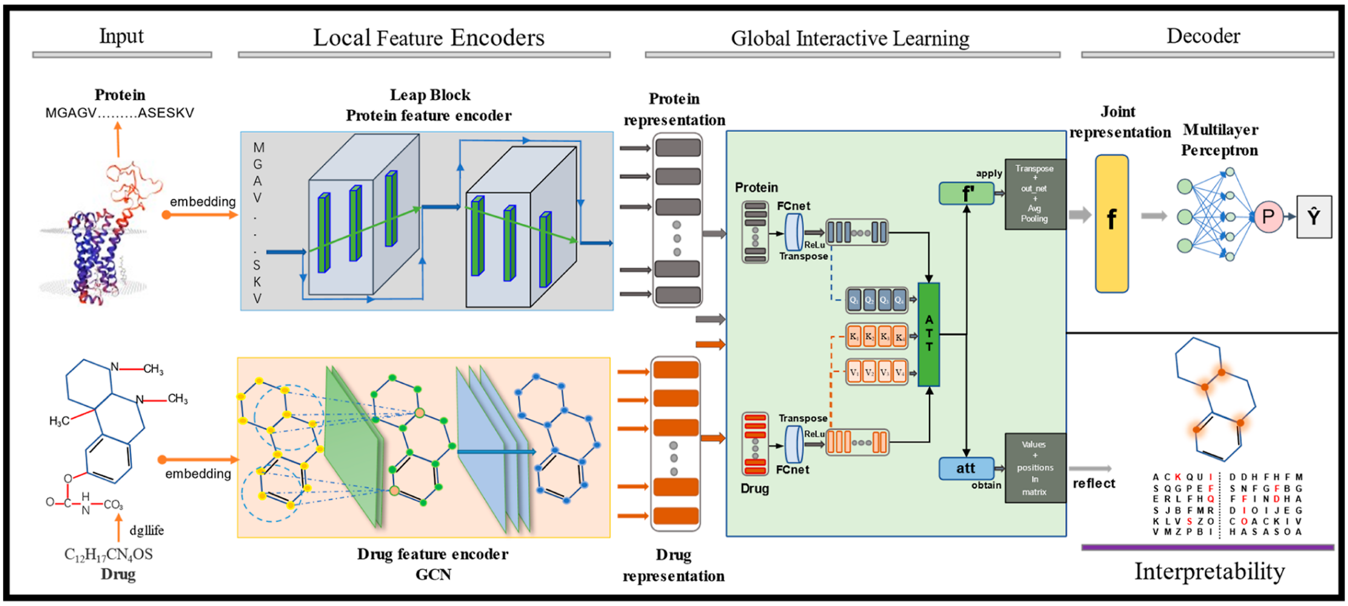

Figure 1. Architecture of MolLoG. The protein sequence is mapped to a feature matrix and then embedded into two leap blocks for encoding. The input drug encoding is transformed into a molecular graph, and the aggregated representation of the neighboring atoms of the drug molecule is encoded through GCNs. The two-modal representation of the obtained drug and protein are input into the GIL module to learn their global interactions, acquiring the joint representation *f* and the attention weight att. The matrix "*f*" is decoded by a multilayer perceptron decoder module to obtain the predicted probability, thus determining the 0/1 label. The matrix "att" will be used for interpretability studies.

Other prominent approaches include methods based on machine learning, such as support vector machine (SVM)[9] and Random Forest (RF),[10] which have their limitations. These include a decrease in computational efficiency when applied to high-dimensional data, the display of an overly sensitive response to parameter modifications, leading to unstable predictive performance, and a lack of adequate interpretability.

As a transformative technique, methods based on deep learning (DL) have made rapid advancements in predicting DTI by integrating multidimensional information into a unified end-to-end framework. These methods have gradually become the optimal solution. Many DL-based models input molecular sequences or graph structures and utilize neural networks to extract latent feature information. Alipanahi et al.[11] employed a deep convolutional neural network (CNN)[12] to predict DNA−protein interactions. The CNN can capture local patterns within drug and protein sequences. Yuan et al. proposed FusionDTA,[13] which uses feed-forward layers and long short-term memory (LSTM) to construct basic blocks of the encoder layer, integrating the intermediaries of drug molecules and proteins into the polymer layer to achieve an output carrier representation of binding affinity. Although methods such as CNNs have improved performance in the sequence and text, graph data, reflecting structural information more effectively, is superior. Nguyen et al. proposed a graph-based model, GraphDTA,[14] which converts drug encoding into an undirected graph with feature graphs and adjacency matrices. The graph convolutional network (GCN),[15] graph attention network (GAT),[16] and graph isomorphism network[17] are designed to extract features from drugs. This is a notable advantage in local feature extraction during the execution of the DTI tasks.

Contrarily, with the protein sequences being considerably longer and their complexity being significantly higher than that of drug encoding, the task of translating these protein sequences into a similar structure represents a challenge. Currently, models that translate only a particular type of protein into graph structures are available, exemplified by GraphProt,[18] which is exclusively designed for RNA-binding proteins. Other approaches include auto covariance, composition transition, and local descriptor.[19] These strategies endeavor to encapsulate some additional fundamental properties of protein sequences. However, these methodologies focus on protein attributes, often overlooking the unique modes of interaction between drug molecules and proteins. This oversight significantly compromises their applicability in the realm of DTI tasks. This issue similarly manifests in methodologies such as ResPRE,[20] which is used for predicting protein contact maps, and DCRNN,[21] used for the prediction of protein secondary structures.

Furthermore, despite the significant progress AlphaFold[22] from DeepMind has made in predicting protein 3D structures, recently generating predictions of 2 billion protein 3D structures from a million species, the reality is that the number of high-accuracy 3D structure proteins only account for a small fraction of known protein sequences. These challenges compel even the most advanced models to continue using methods such as the CNN, RNN,[23] and LSTM[24] to extract protein features from sequences, as seen in models like MolTrans,[25] DeepPurpose,[26] DrugBAN,[27] and MCANet.[28] Although the CNN can extract local aggregated features akin to graph structures, the emergence of the gradient vanishing problem results in some shallow prior knowledge often getting lost with the deepening of the model. Balancing the incorporation of global semantic and intrinsic biological background feature embeddings while considering richer local aggregated features has become a complex issue.

A more pressing challenge is that even if successful predictions of outcomes can be achieved through the captured feature information, without a comprehensive molecular level analysis involving elements such as functional units, residue details, and interaction sites between drugs and their targets,

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the assistance provided for targeted drug development and design is unsatisfactory.[29] However, most of the models mentioned above have not yet surmounted this hurdle.

The key to addressing this issue lies in obtaining importance scores for drugs and targets. Currently, most models score features related to only drugs/targets themselves during the feature extraction stage while neglecting their interaction. In recent years, attention methods have come to the fore as potent tools in identifying key substructures within the predicted targets and ligands of the model. It allows the model to autonomously recognize and focus on the parts most correlated with the prediction outcome, thereby explaining the prediction's provenance. Another significant advantage lies in its ability to facilitate improved joint feature representations within the model, thereby enhancing the predictive performance. Although it can be challenging to achieve interaction learning between the two, some advanced models have nonetheless addressed this issue.

Cutting-edge methods, such as the distance aware graph attention algorithm[30] utilized by Lim et al. and the bilinear attention network of DrugBAN,[27] have made strides in addressing the problem. However, it is evident that in the real-world biological environment, proteins and drugs carry context-related correlation information, with numerous potential interactions existing among different positions. Due to the typically large length of amino acid sequences, the aforementioned model struggles to effectively learn medium to long dependencies and global correlation information. On how to fully utilize the attention mechanism for global joint representation of features and interpret black box results, we made new attempts.

In our research, we introduce a novel model grounded in DL methodologies termed MolLoG. This model is chiefly composed of two modules: local feature encoders (LFE) and global interactive learning (GIL). Utilizing this approach, which smoothly transitions from local to global, we have significantly enhanced DTI prediction and its subsequent interpretability, as visualized in Figure 1. In the LFE module, we employ a GCN combined with a leap block (a technique using residual connections[31]). This pair is tasked with encoding the drug molecular graphs and protein sequences into deep local feature information. These representations are then channeled into the GIL module (an innovative application for the attention mechanism combined with joint interaction of multimodal global features), where they undergo a unique transformation into a global, subblock interaction representation, facilitating a holistic depiction of drug and protein feature interactions.

In summary, our model MolLoG, distinguishes itself from previous research through two key features: (i) the LFE module harmoniously combining local cohesion and holistic associations in feature extraction and (ii) effectively learning multimodal global interaction representations via the GIL module and offering interpretations of predictions utilizing the weighting mechanism from the module.

2. MATERIALS AND METHODS

2.1. Data Sets. Our evaluation involved the assessment of MolLoG, as well as six state-of-the-art baseline models, across four publicly available DTI data sets: BindingDB, BioSNAP, human, and DAVIS. The BindingDB data set is primarily focused on the interactions of small drug-like molecules and proteins.[32] We used the low-bias version of the data set constructed by Bai et al.[33] The BioSNAP data set is a balanced data set created by Huang et al.[25] and Zitnik et al.[34] from the DrugBank database.[35] DAVIS primarily focuses on drug−protein interaction data, applicable in drug discovery and computational chemistry domains.[36] The Human data set was constructed by Liu et al.[37] We used the balanced version of the human data set, containing an equal number of positive and negative samples. Table 1 provides the data statistics for the four data sets.

Table 1. Data Set Statistic

data set	#drugs	#proteins	#interactions
BindingDB	14,643	2623	49,200
BioSNAP	4510	2181	27,465
human	2726	2001	6728
DAVIS	72	382	11,885

2.2. Evaluation Metrics. We utilized five metrics to assess the performance of MolLoG. The AUROC (area under the receiver operating characteristic curve) and AUPRC (area under the precision recall curve) are primary metrics for evaluating model classification performance. The AUROC measures the trade-off between the true positive rate and false positive rate for a classifier at various thresholds. The AUPRC quantifies the trade-off between precision and recall for a classifier at various thresholds. In addition, we also report on the accuracy, sensitivity, and specificity of the optimal F1 score threshold. The definitions of metrics are as follows

$$\text{sensitivity} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{1}$$

$$\text{specificity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3}$$

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{4}$$

where TP represents the number of true positives, TN represents the number of true negatives, FP is the number of false positives, and FN is the number of false negatives. In principle, the higher the value of the above metrics, the better the performance.

2.3. Local Feature Encoders Module. *2.3.1. GCN for Drug Feature Extraction.* For drug compounds, each SMILES string is parsed, with chemical bonds identified, coordinates allocated, and any overlap removed to be converted into a molecular graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$. Each atomic node ($\mathcal{V}$) and chemical bonds ($\mathcal{E}$) are initialized based on their chemical properties. Each atom is represented by a D^a-dimensional integer vector that includes eight pieces of information: implicit hydrogen count, formal charge, total hydrogen count, atom hybridization, unpaired electron count, atom type, atomic degree, and whether the atom is aromatic. This is all implemented via the DGL-LifeSci[38] package. Among the various variants of graph neural networks (GNNs), the GCN extends the approach of neural networks in the representation of graph-structured data, allowing it to update node information based on its neighbor features, own features, and topological data. This computational capacity to reduce dimensions on graph structures meets the computational requirements of predicting similar drugs (targets) based on similar targets (drugs) in the drug target prediction.

We set a maximum allowable number of nodes $\odot^d$. If a molecule has fewer nodes, then we apply zero padding to vacant nodes to convert them into virtual nodes. Consequently, the node feature matrix of each graph is denoted as $M_d \in \mathbb{R}^{\odot^d \times D^a}$. Additionally, we employ a simple linear transformation to define $X_d \in W_D M_d^T$, resulting in a real-valued dense matrix $X_d \in \mathbb{R}^{\odot^d \times D^d}$ that serves as the input feature.

We can represent the ReLU activation function as σ and use $\hat{A}$ and $\hat{D}$ to denote the adjacency matrix and diagonal degree matrix with added self-connections, respectively. $W_d^{(l)}$ represents the weight matrix. Thus, given the X_d, passing it through $(l + 1)$-th layers can be encoded as the feature matrix H_d

$$H_d^{(l+1)} = \sigma(\hat{D}^{-1/2}\hat{A}\hat{D}^{-1/2}H_d^{(l)}W_d^{(l)}),\ H_d^{(1)} = X_d \tag{5}$$

Additionally, the feature of the $(l + 1)$-th node in each layer can be represented as

$$x_i^{(l+1)} = \sigma\left(\sum_{j\in N_i} \frac{1}{C_{ij}} x_i^{(l)} w^{(l)} + b^{(l)}\right) \tag{6}$$

In this context, N_i designates a node and its entire set of neighboring nodes, C_{ij} acts as the normalization factor, $w^{(l)}$ refers to the weight associated with the l-th layer, and $b^{(l)}$ corresponds to the intercept of the same layer.

After processing through the drug encoder, we ultimately obtained multirow feature representations of the drug, collectively referred to as the matrix $O_d \in \mathbb{R}^{\odot^d \times D_{\mathrm{GCN}}}$. D_{GCN} represents the dimension of the final hidden layer in the drug encoder.

2.3.2. Leap Block for Protein Feature Extraction. Each leap block incorporates three CNN layers. The primitive information from protein sequences is amalgamated with the feature output from the initial block and is propagated to the subsequent block. Likewise, the output of each ensuing block integrates the features extracted from the primary trajectory as well as the features derived through the preceding block. This cross-modular data transmission strategy ensures that as we delve into deeper layers, the model persistently unearths profound informational strata and concurrently ingrains a learning mechanism for the already extracted shallow features with globally contextual semantics. And it efficiently tackles the issues of long-distance dependency and gradient disappearance, often encountered in protein encoding.

At the beginning, we perform integer encoding on protein sequences, which encompass 25 characters. This arises from the fact that in addition to the standard 20 amino acids recognized in biology, there are other characteristics employed to denote unique circumstances. Subsequently, all amino acids are initialized into a learnable embedding matrix, denoted as $M_P \in \mathbb{R}^{N \times D^p}$. Herein, N represents the number of amino acid types, while D^p denotes the latent dimension for each type of amino acid. By looking up M_p, each initial protein sequence $p = \{a_1 a_2 ... a_n\}$ that the model needs to learn can be mapped to its corresponding feature matrix $X_p \in \mathbb{R}^{\odot^p \times D^p}$. Here, $\odot^p$ represents the maximum permissible length of the protein sequence, which is set to align different protein lengths for batch training. Protein sequences exceeding the allowable length are trimmed, while those shorter are padded with zeroes.

The protein feature encoder is composed of two leap blocks, each containing three consecutive CNN layers. Each CNN layer encoder H_p is described as follows

$$H_p^{(l+1)} = \sigma(H_p^{(l)}W_p^{(l)} + B_p^{(l)}),\ H_p^{(1)} = X_p \tag{7}$$

where $W_p^{(l)}$ and $B_p^{(l)}$ are the learnable weight filters (matrices) and bias vector in the l-th CNN layer, respectively.

Within the first leap block, protein sequences are perceived as overlapping 3-mer amino acids, illustrated by transforming 'MGAGV...ASESKV' into 'MGA', 'GAG', ..., 'ESK', 'SKV'. Following this, we increase the receptive field to 6 and then to 9, aiming to learn local features of more extended protein sequences. However, persisting with the addition of CNN layers with larger receptive fields might risk diluting some a priori knowledge. Conversely, halting at this stage would leave us oblivious to more nuanced and profound features. To navigate this, we leapfrog the original sequence over the first block and fuse it with the output of the first block, providing the input for the second block. The operation within the second block mirrors that of the first. At the conclusion of each block, we compute and pad the block's output using a 1 × 1 convolution layer to align the dimensions of input and output. Through these two blocks, we extract an ample volume of features, capturing more subtly abstracted deep features while preserving the inherent biological knowledge and contextual semantics of the proteins. The process is formalized as follows

$$BL^{(1)}(X_p) = \mathrm{pad}(F(X_p, H_P^{(3)})) \tag{8}$$

$$BL^{(2)}(X_p) = \mathrm{pad}(BL^{(1)}(X_p)) + X_p \tag{9}$$

$$O_p = BL^{(2)}(BL^{(1)}) + BL^{(1)} \tag{10}$$

In this process, $BL^{(i)}\ (*)$ denotes the workflow of the i-th leap block. In the first block, utilizing the feature matrix X_p derived from the aforementioned mapping, we feed it as input to obtain the output of the third layer of the CNN encoder, as described in (7), wherein each of the three layers in the CNN undergoes normalization, regularization, and processing via the ReLu activation function. $F(*)$ signifies the collection of these operations, while the padding operation pad$(*)$ is performed on the data to maintain consistent input dimensions.

The matrix $O_p \in \mathbb{R}^{\odot^p \times D_{\mathrm{LB}}}$ here (the ultimate output of the protein encoder, D_{LB} represents the number of output channels in the leap block) and the procured O_d (the ultimate output of the drug encoder) are subsequently introduced into the GIL module.

2.4. Global Interactive Learning Module. This module is devised to study the interaction mechanism between drugs and proteins, obtaining their joint representation matrices and attention weight matrices. In our model, drug features are extracted via the graph structure, and protein features are garnered through sequence information. The interaction process involves multimodal feature fusion; hence, we employed an unconventional method. This method is employed at the intersection of computer vision and natural language processing, especially in tasks like visual question answering (QVA),[39] image captioning,[40] and image retrieval.[41] In these tasks, there are usually two types of inputs: visual features (Vf) representing the visual content of the image and

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text query (Tq) denoting the user's query or question about the image. For instance, in the VQA task, "Tq" could be a question like "What color is the cat in the picture (Vf)?". Similarly, in our DTI task, we need to determine the specific segment of the sequence (Tq) that will undergo a reaction with the corresponding atom in the graph structure (Vf). Considering similar challenges in VQA and DTI, we designed the GIL module.

Drug features O_d (Vf) and protein features O_p (Tq) are processed through a custom full connection layer (FCNet), mapping the original features to a new feature space. The purpose of the FCNet is to integrate and abstract the obtained features to achieve a higher-level feature reconstruction. After transposing, we obtain $O_d' \in \mathbb{R}^{D_{\mathrm{FCN}} \times \odot^d}$ and $O_p' \in \mathbb{R}^{D_{\mathrm{FCN}} \times \odot^p}$. D_{FCN} represents the dimension of the last hidden layer of the FCNet. The purpose of transposition is to adjust the two sets of feature matrices to fit the input dimensions of subsequent attention modules. Our objective is to achieve global interaction learning between each molecule of the drug and protein, rather than just interacting certain features of multiple molecules, which would be meaningless.

Afterward, these two new types of features are then fed into an interaction module composed of multihead attention. Under typical circumstances, the core idea of the attention mechanism revolves around calculation based on the query (Q), key (K), and value (V). Each input is correlated with the generation of the respective Q, K, and V. Subsequently, the similarity between Q and K is computed to generate a weight distribution, which is then used to perform a weighted summation of V to yield the output. In our work, the features of the drug are encoded to simultaneously generate K and V. This is due to our aim to seek the most relevant information in K (generated by O_d') via Q (generated by O_p'), and then generate the answer using the corresponding V (also generated by O_d'). This technique of formulating keys, values, and queries from varying sources becomes particularly effective when the input data display inherent relationships or autocorrelations. As seen in our DTI task, after undergoing the custom FCNet abstraction, where drug and protein features demonstrate strong relatedness, this results in richer representations being learned.

Due to the involvement of multiple linear transformations and transpositions, various matrices undergo multiple changes in dimensions during this process. To enhance the clarity of our method, we can express it mathematically as follows.

We set the number of heads for parallel computation as h (must be divisible by h). For each head$_i$ ($1 \le i \le h$), we can define three weight matrices, $W_q^i \in \mathbb{R}^{L_Q \times D_{\mathrm{FCN}}}$, $W_k^i \in \mathbb{R}^{L_K \times D_{\mathrm{FCN}}}$, and $W_v^i \in \mathbb{R}^{L_V \times D_{\mathrm{FCN}}}$, which will be used for linear transformations. Here, L_Q, L_K, and L_V denote the lengths of the Q, K, and V vectors, respectively. In general, these three values are equal. Given O_d' and O_p', the symbol $\cdot$ denotes matrix multiplication. We can calculate the matrices Q_i, K_i, and V_i

$$Q_i = W_q^i \cdot O_p', \quad K_i = W_k^i \cdot O_d', \quad V_i = W_v^i \cdot O_d' \tag{11}$$

With the preparations made, we proceeded to calculate the attention weights for each head (att$_i$). These attention weights are then applied to the values (V_i), resulting in the output matrix for each head (head$_i$).

$$\mathrm{att}_i = \mathrm{softmax}\left(\frac{Q_i \cdot K_i^T}{\sqrt{d_i^k}}\right) \tag{12}$$

$$\mathrm{head}_i = \mathrm{softmax}\left(\frac{Q_i \cdot K_i^T}{\sqrt{d_i^k}}\right) \cdot V_i \tag{13}$$

In this context, d_i^k represents the dimension of the key vector in head i. $\sqrt{d_i^k}$ is used for scaling the attention weights, which helps stabilize the training process and prevent gradient explosion or vanishing due to large dimensions of queries and keys. The softmax function is applied to normalize the scores, converting them to a probability distribution. Next, we concatenate the attention weights matrix M_{att} of each head.

$$M_{\mathrm{att}} = \mathrm{concat}(\mathrm{att}_1 \ldots \mathrm{att}_h) \cdot W_{\mathrm{att}} \tag{14}$$

Here, $W_{\mathrm{att}} \in R^{D_{\mathrm{FCN}} \times D_{\mathrm{FCN}}}$. M_{att} is an important part of interpretable analysis. For the matrix head$_i$, we also concatenate it to obtain the joint representation f'

$$M_{f'} = \mathrm{concat}(\mathrm{head}_1 \ldots \mathrm{head}_h) \cdot W_0 \tag{15}$$

Here, $W_0 \in R^{D_{\mathrm{FCN}} \times D_{\mathrm{FCN}}}$. Meanwhile, we apply the following three operations to f' in order to obtain the final joint representation f, which is used for decoding.

1. Transpose it again to match the dimensions of the initial feature map.
2. Apply average pooling, which helps reduce the spatial size of the input data and smooth out the overall features.
3. Apply another fully connected layer (out_net) to generate the final joint representation.

Overall, the three steps we have taken are all dedicated to further strengthening the interaction of the features we extract. Moreover, the unique mechanism of the GIL module greatly facilitates the integration of drug features and protein features, which in turn enhances the predictive performance of the model.

2.5. Molecular Level Interpretability Analysis. We load the attention weight matrix derived from the GIL module, where each value in the matrix represents the importance of each drug atom to each target molecule. We also acquire the row and column positions of points in the attention weight matrix that exceed a certain threshold within the entire matrix and side information (the corresponding protein representation and drug representation). Given that typically over 20% of the atoms in drugs constitute active functional groups, we have set this threshold at 80%, aiming to identify the top 20% most important atoms.

For molecular elucidation of drugs, we load SMILES encoding and create molecular graphs through the RDKit[42] toolkit, obtaining the count of actual atoms in the molecule and corresponding attention weights. GCNs propagate and aggregate node feature information across layers. In each layer, the feature vectors of nodes are updated while maintaining their positional indices in the feature matrix. In the process of drawing the molecular image, atoms highlighted beyond a certain threshold are depicted in a vibrant orange-red color (E94E4E), representing those atoms in the drug that are expected to participate primarily in the reaction. For protein analysis, we consider each column of M_{att} as an attention

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weight vector, mapping this weight vector onto the one-dimensional sequence of the protein and thereby accentuating those amino acid residues that are deemed important by the model. Meanwhile, we can still localize individual protein fragments. This is due to the sliding window[43,44] property of the CNN within the leap block.

Visualization of the three-dimensional structure is achieved using the PDB[45] Web site and Pymol.[46] Here, merely the method is described; more intriguing cases and biological explanations are provided in Section 3.6. Note that in this paper, our main focus is on the interpretability of drug atoms. As for the interpretation of targets, we provide necessary supplements and explanations within the text.

3. RESULTS AND DISCUSSION

3.1. Implementation and Hyperparameters. Our experimental setup was deployed on a system equipped with two RTX 2080 Ti units, each with specifications of 11 GB of graphics memory, 12 virtual CPUs, and 40 GB of RAM. The execution environment and versions of tools adopted by MolLoG include the following: PyTorch 1.11.0 + CUDA 11.3, Python 3.8.10, DGL 0.9.1post1, Numpy 1.23.5, Pandas 1.5.3, and RDKit 2022.09.5. Our model comprises a substantial quantity of hyperparameters, with key parameters highlighted in Table 2.

Table 2. Hyperparameter Used in MolLoG

hyperparameter	setting
optimizer	adam
learning rate	5×10^{-5}
MAX_Epoch	100
BATCH_SIZE	64
leap block	2
CNN Kernel size	[3, 6, 9]
GCN dimensions	[128, 128, 128]
heads of attention	4
attention pooling size	3

3.2. Evaluation of Prediction Performance. Each experimental data set was randomly split into a training set, a validation set, and a test set, maintaining a 7:1:2 ratio. The test set was considered "unknown" for assessment purposes. For each data set, we performed five independent iterations with varied random seeds. The model exhibiting the best AUROC on the validation set was deemed the top-performing model, which was subsequently assessed on the test set.

Here, we compare MolLoG against six baselines under a random split setting: SVM,[9] RF,[10] DeepConv-DTI,[47] GraphDTA,[14] MolTrans,[25] and DrugBAN.[27] The performance of MolLoG on the BindingDB and BioSNAP data sets is presented in Table 3. Bold data represent the optimal values in the table. MolLoG consistently outperforms the baselines in terms of the AUROC, AUPRC, and accuracy and is highly competitive in specificity.

Due to some unique characteristics of the human and DAVIS data sets, we investigated the model's performance on these two data sets separately later.

3.3. Using the Cold Pair Split Strategy on the Human Data Set. Under the standard random split paradigm, DL-based models displayed stable and commendable performance on the human data set (AUROC > 0.97, fluctuation < 0.005). Among these, only MolLoG and DrugBAN achieved an AUROC exceeding 0.98. However, Chen et al. highlighted a potential hidden ligand bias inherent in the human data set, proposing that models trained on this data set may primarily learn drug-specific features rather than the desired interaction patterns.[48] Therefore, we employed a cold pair split strategy for a more rigorous evaluation of our model. This involved randomly assigning 5% and 10% of DTI pairs to validation and test sets, respectively, followed by the removal of the corresponding drugs and proteins from the training set. This strategy ensures the segregation of the data set based on drug and protein independence, leading to a scenario where drugs and proteins present in the test set are entirely unseen during training.

Figure 2 showcases a significant dip in the performance of all models under this stringent strategy, with traditional models such as SVM and RF showing pronounced degradation. Despite this challenging scenario, MolLoG manages to deliver superior results when compared to other state-of-the-art DL baselines. We additionally applied the cold split strategy to the remaining three data sets. All models experienced a decrease in performance across all data sets. The key metrics AUROC and

Table 3. Performance Comparison on the BindingDB and BioSNAP Data Sets With Random Split

method	AUROC	AUPRC	accuracy	sensitivity	specificity
			BindingDB		
SVM	0.939 ± 0.001	0.928 ± 0.002	0.825 ± 0.004	0.781 ± 0.014	0.886 ± 0.012
RF	0.942 ± 0.011	0.921 ± 0.016	0.880 ± 0.012	0.875 ± 0.023	0.892 ± 0.020
DeepConv-DTI(2019)	0.945 ± 0.002	0.925 ± 0.005	0.882 ± 0.007	0.873 ± 0.018	0.894 ± 0.009
GraphDTA(2021)	0.951 ± 0.002	0.934 ± 0.002	0.888 ± 0.005	0.882 ± 0.012	0.897 ± 0.008
MolTrans(2021)	0.952 ± 0.002	0.936 ± 0.001	0.887 ± 0.006	0.877 ± 0.016	0.902 ± 0.009
DrugBAN(2023)	0.953 ± 0.005	0.943 ± 0.003	0.896 ± 0.005	0.887 ± 0.012	0.906 ± 0.007
MolLoG	**0.962 ± 0.004**	**0.951 ± 0.006**	**0.905 ± 0.005**	**0.898 ± 0.012**	**0.912 ± 0.005**
			BioSNAP		
SVM	0.862 ± 0.007	0.864 ± 0.004	0.777 ± 0.011	0.711 ± 0.042	0.841 ± 0.028
RF	0.860 ± 0.005	0.886 ± 0.005	0.804 ± 0.005	0.823 ± 0.032	0.786 ± 0.025
DeepConv-DTI(2019)	0.886 ± 0.006	0.890 ± 0.006	0.805 ± 0.009	0.760 ± 0.029	0.851 ± 0.013
GraphDTA(2021)	0.887 ± 0.008	0.890 ± 0.007	0.800 ± 0.007	0.745 ± 0.032	0.854 ± 0.025
MolTrans(2021)	0.887 ± 0.002	0.891 ± 0.005	0.802 ± 0.004	0.755 ± 0.021	0.843 ± 0.023
DrugBAN(2023)	0.896 ± 0.005	0.900 ± 0.004	**0.834 ± 0.008**	**0.828 ± 0.021**	0.847 ± 0.010
MolLoG	**0.898 ± 0.003**	**0.902 ± 0.005**	0.832 ± 0.006	0.826 ± 0.027	**0.856 ± 0.022**

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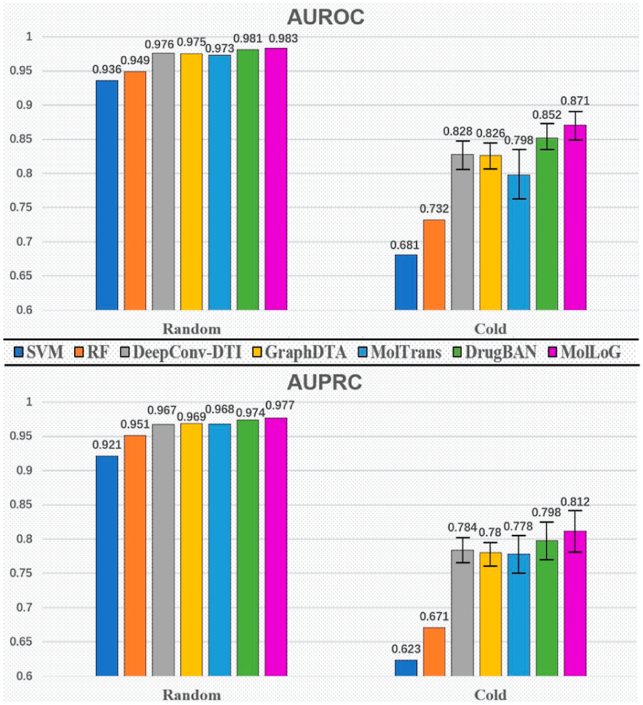

Figure 2. Performance on the human data set. The numbers in the graph represent the average values, while the error bars represent the performance fluctuations.

AUPRC for MolLoG decreased by approximately 0.02 in the BingingDB data set, around 0.02 in the BioSNAP data set, and roughly 0.04 in the DAVIS data set. However, MolLoG still outperforms other models in terms of overall performance.

3.4. Small Sample Data Set Validates Generalization. To further assess whether the model possesses good generalization capabilities when dealing with a larger number of parameters and modules, we conducted experiments using the DAVIS data set under random split conditions.

The DAVIS data set encompasses experimentally validated drug−protein interaction data. Nevertheless, they contain only a limited number of drug and protein samples. To expedite researchers in extracting requisite drug−protein interaction information for analysis and modeling, the data set's structure is relatively simplistic. This presents a challenge in balancing the generalization capability of the model, the number of parameters, and the complexity.

Consequently, we have discarded relatively simplistic models, such as SVM and RF.

From Table 4, we can discern that MolLoG, through its unique learning method from local to global, exhibits commendable generalization capabilities, even when faced with a data set with a relatively simplistic structure and fewer samples, despite its intricate hierarchical structure and a greater number of parameters. Notably, there is a significant improvement in two crucial metrics: accuracy and specificity.

3.5. Ablation Study. We have conducted three ablation studies to investigate the impact of the LFE module and the GIL module on MolLoG. This study specifically includes the following aspects:

1. The impact of using different GNN variants to extract drug features on the performance of the model;
2. The impact of using the leap block, as well as the number of blocks and layers, on protein feature extraction;
3. The effect of using the GIL module to globally integrate the data representation and the impact of the number of attention heads in the module on performance.

In Table 5, we present the performance of the best round in terms of the F1 score from five rounds of experiments. The

Table 5. Three Kinds of Ablation Studies

ablation tests	AUROC	AUPRC	accuracy	sensitivity	specificity
Part 1. The Impact of Using Different GNN Variants for Drug Feature Extraction under the DAVIS Data Set					
GAT	0.8362	0.8125	0.7592	0.7864	0.7879
GraphSAGE	0.8329	0.8078	0.7842	0.7869	0.7633
MolLoG (GCN)	**0.8574**	**0.8592**	**0.7996**	**0.8011**	**0.8010**
Part 2. The Impact of the Use of the Leap Block and the Number of Blocks and Layers on the Results under the DAVIS Data Set					
only 3 CNNs	0.8361	0.8022	0.7556	0.7626	0.7636
only 6 CNNs	0.8311	0.7918	0.7501	0.7523	0.7816
3 blocks with 3 CNNs	0.8512	0.8488	0.7826	0.7820	0.8028
2 blocks with 4 CNNs	**0.8663**	**0.8592**	**0.7847**	**0.7889**	**0.8055**
Part 3. The Impact of Using the GIL Module to Globally Integrate Data Representation on the Results under the BioSNAP Data Set					
not used	0.8741	0.8782	0.8045	0.7964	0.8126
used	**0.9011**	**0.8971**	**0.8282**	**0.8257**	**0.8473**

experiments indicate that each module we have employed provides performance enhancement for MolLoG. During a particular part of the experiment, all other modules remain consistent with MolLoG.

In part 1, we compared the GAT that uses the attention mechanism and GraphSAGE,[49] which introduces a learnable aggregation function that allows nodes to weight different neighboring nodes during aggregation. The experiment indicated that the GCN that we employed was the most effective.

In part 2, we continued to expand the receptive field based on a 3-layer CNN. The introduction of more parameters did not enhance the model's performance; instead, there was a decline, indicating the gradient vanishing issue mentioned earlier. When we added the leap block, all metrics saw a

Table 4. Performance Comparison on the DAVIS Data Sets With Random Split

method	AUROC	AUPRC	accuracy	sensitivity	specificity
DAVIS					
DeepConvDTI(2019)	0.796 ± 0.023	0.788 ± 0.036	0.724 ± 0.008	0.694 ± 0.033	0.721 ± 0.013
DeepDTA(2021)	0.801 ± 0.016	0.790 ± 0.038	0.731 ± 0.013	0.735 ± 0.039	0.720 ± 0.024
MolTrans(2021)	0.828 ± 0.018	0.796 ± 0.028	0.733 ± 0.015	0.702 ± 0.043	0:803 ±0.052
DrugBAN(2023)	0.814 ± 0.012	0.825 ± 0.033	0.743 ± 0.015	0.748 ± 0.034	0.776 ± 0.019
MolLoG	**0.852 ± 0.005**	**0.856 ± 0.029**	**0.782 ± 0.011**	**0.766 ± 0.028**	**0.808 ± 0.011**

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significant increase. Note that in MolLoG, we use 2 blocks with 3 CNNs because it offers better robustness and generalization.

In part 3, If we do not use the GIL module to abstract and globally interact features but instead concatenated the features directly, there would be a significant performance decline.

We also compared the effect of different numbers of attention heads on performance. Under the BindingDB data set, we plotted the AUROC and AUPRC of the model on the validation set for the first 60 epochs as a line chart, as shown in Figure 3: the model's performance demonstrates a pronounced ascending trend in response to an increase in the number of training epochs. Among these, the model exhibits optimal performance when the number of heads in the multihead attention module is set to four.

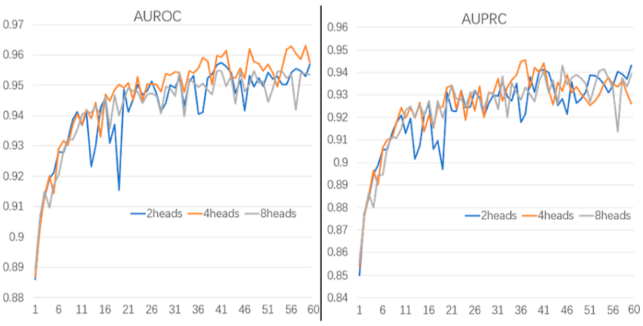

Figure 3. Line chart of the AUROC and AUPRC for the first 60 epochs on the validation set under the BindingDB data set.

In summary, this validates the effectiveness of our strategy, which involves extracting local features and facilitating global interactions.

3.6. Case Study. *3.6.1. DTI Prediction.* To assess the predictive efficiency of MolLoG, we embarked on a case study analysis involving drugs D00235 (Atenolol[50]) and D00567 (Celecoxib[51]). For both drugs, we manually scrutinized the top 10 predicted targets, corroborating our findings with data obtained from publicly accessible databases, which led to full verification (Table 6). Considering these promising results, we are motivated to speculate that several high-scoring interactive targets, although currently unconfirmed, could potentially exhibit actual interactions with the drugs under study. This will facilitate the development and refinement of new drugs.

3.6.2. Interpretability at the Molecular Level. We have selected 3U5J (first bromodomain of human BRD4 in complex with alprazolam) and 5XPR (human endothelin receptor type-B in complex with the antagonist bosentan) for a study of interpretability at the molecular level. These two cocrystal ligands from the Protein Data Bank have resolutions of 1.60 and 3.60 Å respectively, and neither is part of the model training set. The drugs alprazolam and bosentan play critical roles in the treatment of specific diseases and have become indispensable for many patients in terms of improving the quality of life and survival.

Alprazolam, a triazolo benzodiazepine with an intermediate onset, is commonly administered for the treatment of panic disorders and generalized anxiety, as well as anxiety concomitant with depression.[52] In the complex reaction between alprazolam and BRD4, we successfully predicted some of the critical atoms involved in the alprazolam reaction (Figure 4B). Our model explained that the highlighted atoms would bind to I146, L94, and Y139. When analyzing proteins, our model failed to accurately identify these three targets. In the BRD4 carrier, the acetyllysine in alprazolam coordinates with the conserved tyrosine residue Y97, a point MolLoG successfully identified. Co-crystal analysis revealed the triazole ring acting as an acetyllysine mimic scaffold, forming a hydrogen bond with the conserved residue N140. Regrettably, we did not capture the fact that N140 serves as a hydrogen bond anchor point for the acetylated substrate. A literature study suggested that the likely reason for this was the absence of a second hydrogen bond with N140 in the alprazolam complex, which probably resulted in lower ligand affinity.[53]

In Figure 4B, the green atom was initially labeled by MolLoG. However, when we substituted the experimental drug with triazolam (which has only one additional chlorine atom at the green atom compared to alprazolam), MolLoG removed the label. This is because the substitution of chlorine substitution at the 2-position of the phenyl ring is intolerant. MolLoG captures this subtle change.

Bosentan, a dual endothelin receptor antagonist, is employed in the management of pulmonary arterial hypertension.[54] For the structurally complex 5XPR, our model likewise successfully identified key interaction sites between bosentan and the endothelin receptor type-B (ETB) (refer to Figure 4F), Bosentan is composed of multiple aromatic groups, structured around a central pyrimidine template and four substituents: 2-pyrimidyl, 4-sulfonamide, 5-phenol, and 6-ethylene glycol. We were able to predict several of the interactions noted within the high-resolution structure (Figure 4G). The sulfonamide engages in ionic interactions with Lys182$^{3.33}$ and Lys273$^{5.38}$; the 4-*t*-butylphenyl, associated with the sulfonamide, contacts Val177$^{3.28}$ and Pro178$^{3.29}$; the pyrimidine at the 2-position of the central pyrimidine forms a hydrophobic contact with Ile372$^{7.39}$; and the methoxyphenoxy group ether moiety of 5-phenol reacts with the Val185$^{3.36}$ and Leu277$^{5.42}$ residues in the

Table 6. Top 10 Prediction Target List for Drugs "D00235" and "D00567"

drug ID	target ID	prediction	evidence	drug ID	target ID	prediction	evidence
D00235	P18090	0.932	KEGG	D00567	P23280	0.774	KEGG
D00235	Ch_38603	0.921	KEGG	D00567	P00918	0.968	DrugBank
D00235	Ch_735421	0.745	KEGG	D00567	P23219	0.862	KEGG
D00235	P28564	0.862	KEGG	D00567	P35454	0.813	DrugBank
D00235	Q92887	0.843	KEGG	D00567	P07451	0.719	DrugBank
D00235	P07550	0.910	DrugBank	D00567	Ch_107983	0.822	KEGG
D00235	O95342	0.861	DrugBank	D00567	Ch_662815	0.855	KEGG
D00235	P08588	0.793	DrugBank	D00567	O15530	0.842	DrugBank
D00235	P61169	0.878	KEGG	D00567	O76074	0.789	KEGG
D00235	P07550	0.746	DrugBank	D00567	EBI_101485	0.831	KEGG

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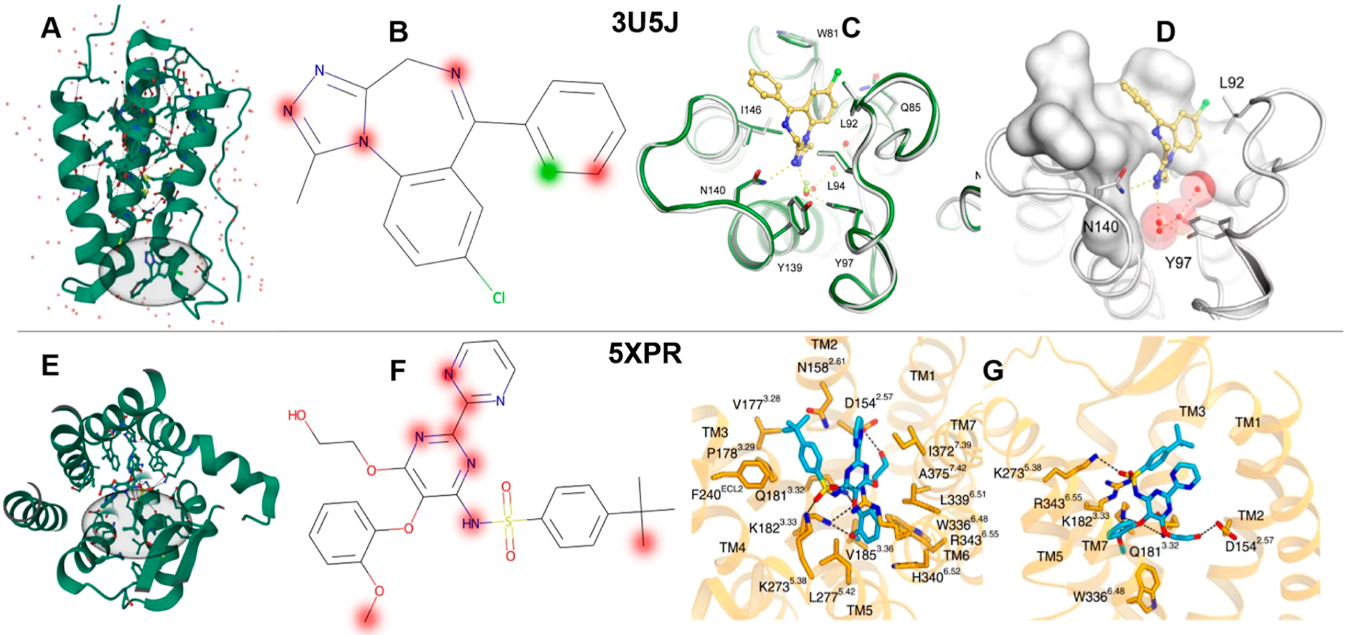

Figure 4. Important visualization and supplementation of biological and chemical knowledge for the prediction results. (A,E) Overview of two complex structures. The black semitransparent ellipses focus on the drug and ligand reaction areas. (B,F) Drug atoms that MolLoG predicts to be key in the reactions in both cases. (C) Overlay of the BRD4 apolipoprotein structure (green stripe-like structure) and alprazolam ligand complex (rod-like structure).[54] This figure highlights the main interaction residues and the details of the alprazolam binding site. (D) Detailed depiction of the cocrystal ligand. The alprazolam binding site is represented by a transparent white surface. (G) Overlay of bosentan (rod-like structure) and ETB (in pale yellow).

transmembrane core. Additionally, there were low-weight prediction indications on the 6-ethylene glycol, potentially because the hydroxyl moiety of bosentan, carrying ethyleneglycol, solely forms a hydrogen bond, leading to a weakened receptor interaction.[55] In the localization of protein residues, we identified only three targets, labeled as Lys182$^{3.33}$, Pro178$^{3.29}$, and Val185$^{3.36}$.

Overall, given that our model is trained solely on one-dimensional protein sequences, it can result in predictive omissions and mismatches with the atoms of the drug. The task of interpreting protein structures, a notably challenging endeavor, demonstrates the constraints of our model. Conversely, there has been a notable improvement in the interpretability of other prediction outcomes. MolLoG can even predict certain biochemical reactions that have been verified to occur only under specific circumstances. In addition, given that the interpretability provided by MolLoG is learned intrinsically from the vast DTI data, such an interpretation might uncover hitherto unexplored hidden insights. This feature could potentially provide researchers with a directional approach in conducting high-cost biochemical experiments.

3.7. Limitations and Future Work. In the data preprocessing stage, if a scientifically effective method emerges to encode proteins into 2D graph structures similar to drugs, applying it to MolLoG could significantly improve the prediction accuracy and interpretability. Additionally, integrating quantum mechanical features[56] describing DTIs into feature extraction would also be an interesting direction as this approach can achieve extremely high precision. However, the cost of acquiring the data for this method may be high. In terms of interpretability, accurately identifying the original fragments becomes challenging. While the incorporation of positional encoding could establish corresponding relationships, it implies a tremendous performance cost. In our preliminary experiments, introducing positional encoding increased the model training time by dozens of times. Hence, integrating positional encoding and applying knowledge distillation[13,57] to reduce parameter quantity could potentially be an effective direction.

4. CONCLUSIONS

As a powerful supplement to in vitro experiments, in the era of big data and artificial intelligence, we propose a DL model MolLoG, which can learn and predict potential DTIs on a large scale, thereby accelerating the drug development process in an efficient and economical manner. MolLoG is primarily composed of “LFE” and ‘GIL’ modules. Experimental results show that through its learning approach from local to global, it efficiently extracts local features and delivers effective global joint representations, outperforming the current state-of-the-art models. Building on this foundation, we map the attention weights, which integrate two modalities, onto the subsequences of proteins and the atoms of drug compounds. This approach allows us to achieve visualization of results at the molecular level. This enhanced visual representation can provide deeper biological insights and more targeted recommendations for practical medical experiments.

■ ASSOCIATED CONTENT

Data Availability Statement

The source code and implementation details of MolLoG are freely available at the GitHub repository: https://github.com/fbbgood/MolLoG.git.

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